



SESSION 1

HELLO, PYTHON

Information Systems Development (SE)

Dr. Pejman Ebrahimi · University of Liechtenstein

TODAY IN FOUR MOVES

01

Text and
numbers

02

A Mad Libs
generator

03

HackerRank
problems

04

Claude as
your tutor

HOW EVERY SESSION RUNS

0 : 00

Recap and a warm-up puzzle

0 : 20

A new idea, live-coded together

1 : 10

The main build of the day

1 : 55

HackerRank. No AI in this block.

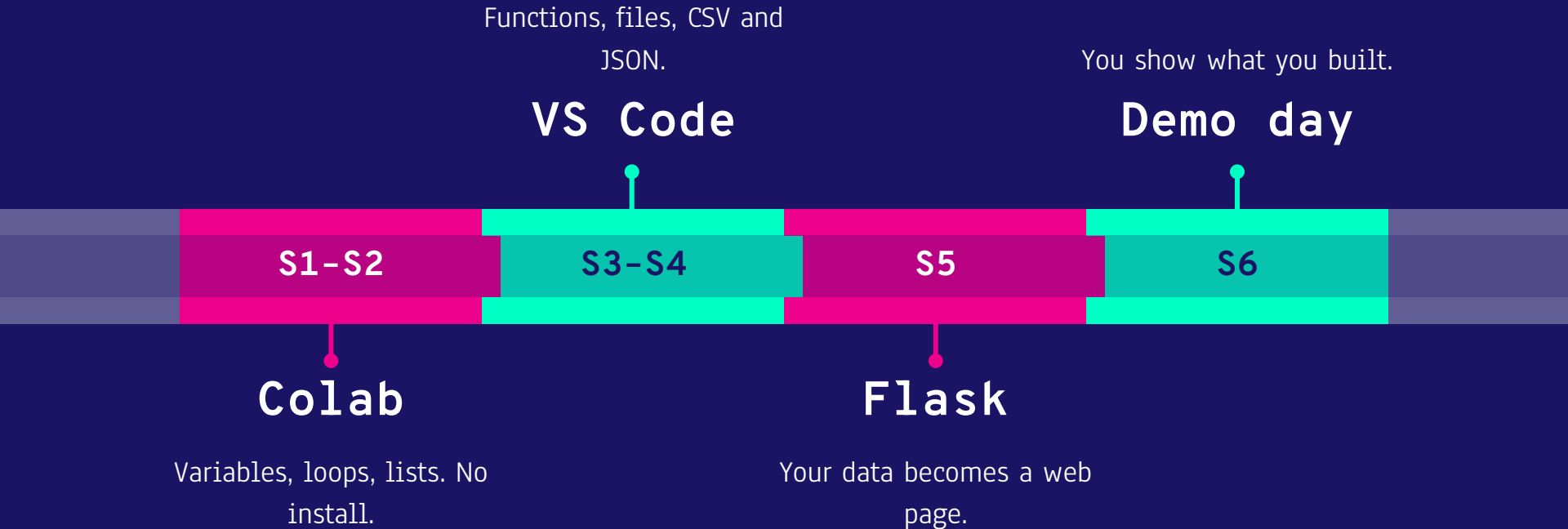
2 : 20

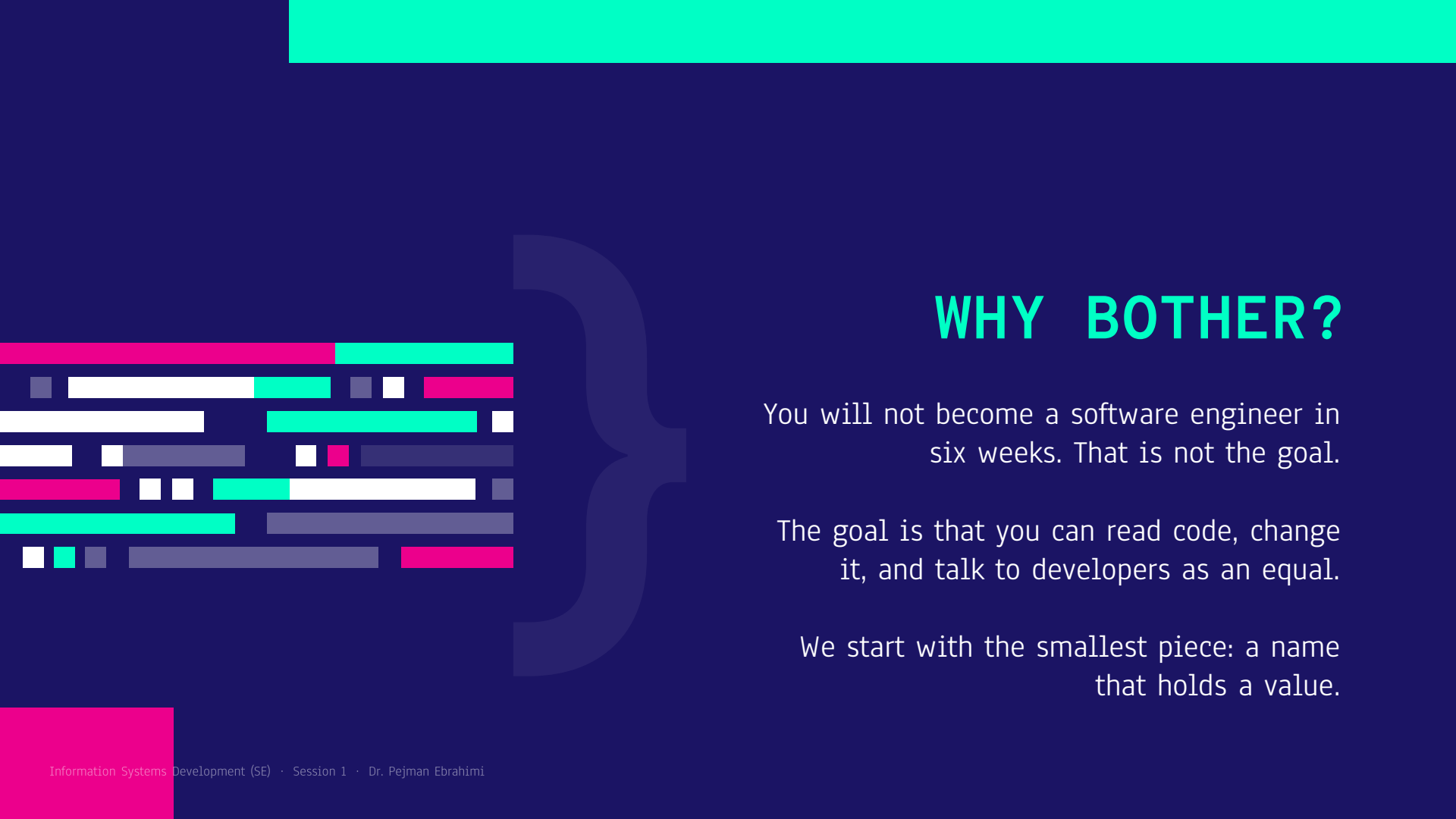
AI block. One set exercise.

2 : 45

Wrap-up, homework, next week

THE COURSE IN FOUR MOVES





WHY BOTHER?

You will not become a software engineer in six weeks. That is not the goal.

The goal is that you can read code, change it, and talk to developers as an equal.

We start with the smallest piece: a name that holds a value.

FOUR HOUSE RULES



Errors are normal

Information, not judgement.



Attempts, not scores

HackerRank counts submissions only.

Nothing to install

Colab runs in your browser.



One AI-free zone

No AI during HackerRank.
After, yes.





01

GOOGLE COLAB

WHAT IS COLAB

A Python notebook that runs on Google's computers, inside your browser.

A notebook is a stack of cells. Write code in a cell, press Shift + Enter.

The result appears under the cell.



THE ONE COLAB TRAP

The trap

You run cells out of order,
so a variable vanishes.



The fix

Runtime → Restart and run
all.

Then read from the top down.



SWITCH TO COLAB
AND TYPE ALONG





02

VARIABLES AND TYPES

THREE TYPES FOR TODAY

str

Text, in quotes. "Laptop", "Anna"



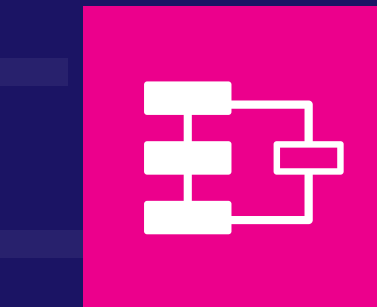
int

Whole numbers. 899, 4, 0



float

Numbers with a decimal. 8.99, 0.5



VARIABLES IN ACTION

```
# one product in our shop
name  = "Laptop"
price = 899.0
stock = 4

# "=" means: put this value
# into that name

print(name)      # Laptop
print(stock)     # 4
```

PRINTING, TWO WAYS

Clumsy

```
print("Price: "  
      + str(price))
```

f-string

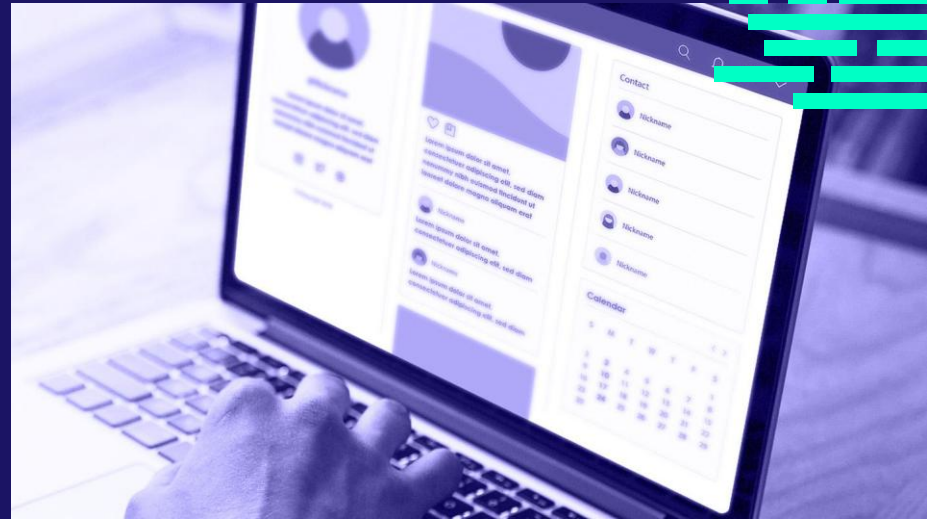
```
print(f"Price:  
      {price} EUR")
```



QUICK CHECK

```
price = 899  
print("price")
```

What appears?





03

TALKING TO THE USER

ASKING THE USER

```
name = input("Your name: ")
```

```
print(f"Hello, {name}!")
```

```
# input() stops and waits.
```

```
# It hands back whatever
```

```
# the person typed.
```

input() IS ALWAYS TEXT

This breaks

```
age = input("Age: ")  
age + 1 → TypeError
```



This works

```
age = int(input("Age: "))  
age + 1 → 31
```



**YOUR TURN
RUN IT, THEN
BREAK IT**



THREE ERRORS TODAY

NameError

A name Python has never seen.



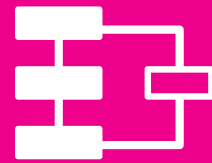
TypeError

You mixed text and numbers.



SyntaxError

A quote or bracket is missing.



READ IT FROM THE BOTTOM

Traceback

A wall of red text.

Start at the bottom.

Last line

The error type and message

Line above it

The line number in your code

Above that

The line that actually broke

Then fix

Change one thing. Run again.



04

MAD LIBS

MAD LIBS: THE PLAN



1. Collect

Ask a classmate for a noun, a verb and an adjective. Do not show them the story.



2. Assemble

Drop the words into a story with an f-string, print it, and read the best one out loud.

MAD LIBS STARTER

```
noun = input("A noun: ")  
verb = input("A verb: ")  
adj  = input("An adjective: ")
```

```
print(f"Our {adj} shop sells")  
print(f"exactly one {noun}.")  
print(f"Customers {verb}!")
```

```
# now add three more words
```



05

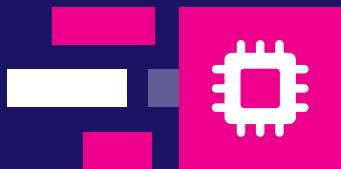
HACKERRANK

HOW HACKERRANK WORKS



Input arrives

Read it with `input()`. Never typed in.



Leave the rest alone

Code below the stub already works.

Sometimes a stub

If you see `def`, you return.



Run before Submit

A red test is information.



TODAY'S THREE PROBLEMS

	Stream	It teaches	Time
Hello, World!	A	The platform itself	5 min
Arithmetic Operators	A	Reading numbers in	10 min
Python: Division	A	/ versus //	10 min

DIVISION: / VERSUS //

10 / 3

3.333333333333335
Always a float.



10 // 3

3
The whole part only.



100%

of the HackerRank marks come from submitting an attempt — rank is worth zero.



06

CLAUDE AS A TUTOR

THE AI RAMP, IN ORDER

Tutor

Sessions 1–2. It explains, you write.



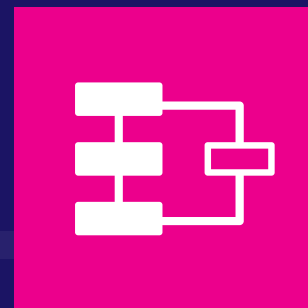
Partner

Sessions 3–4. It reviews your code.



Builder

Sessions 5–6. It scaffolds, you decide.



THE PROMPT FOR TODAY

I am a beginner learning Python.

Here is my error:

<paste the whole red message>

Explain what it means in simple terms.

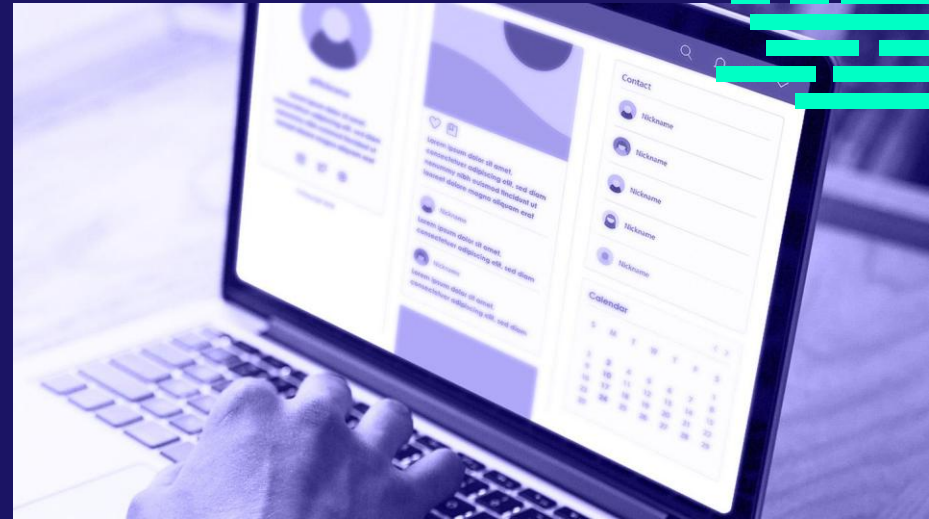
Do NOT give me the corrected code yet.

YOUR TASK

Three snippets in the notebook. Each one breaks.

Run it, copy the red message, ask Claude why.

Then fix it yourself.





“Ask why, not what. A tool that
hands you the answer teaches
you nothing.”

— THE RULE OF SESSION 1

HOMEWORK

1. A script that introduces you: name, age, one favourite thing, printed with f-strings.
2. One paragraph: an error you hit today, and how you got past it.

Next week: loops, and a guessing game.





Thanks !

Questions?

pejman.ebrahimi@uni.li

Next: Session 2

Decisions and Repetition